

Valtti Järvi

COMPARATIVE STUDY OF FOUNDATIONAL TIME SERIES MODELS FOR EMERGENCY DEPARTMENT OCCUPANCY FORECASTING

Bachelor's Thesis
Faculty of Information Technology and Communication Sciences
Examiner: M.Sc. Eetu Pulkkinen
December 2025

ABSTRACT

Valtti Järvi: Comparative study of foundational time series models for emergency department occupancy forecasting

Bachelor's Thesis

Tampere University

Bachelor's Programme in Computing and Electrical Engineering

December 2025

Emergency department overcrowding is a significant threat to patient safety..

Keywords: time series forecasting, foundational models, emergency department occupancy prediction

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TIIVISTELMÄ

Valtti Järvi:
Kandidaatin tutkielma
Tampereen yliopisto
Tieto- ja sähkötekniikan kandidaattiohjelma
Joulukuu 2025

Avainsanat: keyword, keyword, keyword, keyword, keyword

Tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck -ohjelmalla.

CONTENTS

1.	Introduction	1
2.	Background	2
2.1	Time series forecasting	2
2.2	Foundational models	2
2.3	Related work	2
2.3.1	Current state of art in ED forecasting	2
2.3.2	Performance of foundational models in other fields	2
3.	Methodology	3
3.1	Models	3
3.2	Data	3
3.3	Experimental setup	3
3.4	Evaluation	3
3.5	Finetuning	3
4.	Results	4
4.1	Main results	4
4.2	Effect of context length	4
4.3	Effect of covariates	4
4.4	Visualizations	4
5.	Discussion	5
6.	Conclusion	6
7.	References	7

LIST OF SYMBOLS AND ABBREVIATIONS

a	kiihtyvyys
CC-lisenssi	Creative Commons -lisenssi
F	voima
ISO	Kansainvälinen standardointiorganisaatio
$\text{\LaTeX}$	ladontajärjestelmä tieteelliseen kirjoittamiseen
m	massa
$\mathbb{R}$	reaaliluvut
SI-järjestelmä	kansainvälinen mittayksikköjärjestelmä (ransk. <i>Système international d'unités</i>)
TAU	Tampereen yliopisto (engl. Tampere University)
TUNI	Tampereen korkeakouluyhteisö (engl. Tampere Universities)
URL	verkkosivun osoite (engl. Uniform Resource Locator)

1. INTRODUCTION

Overcrowding of hospital emergency departments is harmful to patients.(lähde)

Emergence of foundational time series models raises questions about how well they perform on unseen data.

In this thesis foundational time series models are evaluated using ED occupancy data from . Performance of these models are compared to machine learning models that have been trained and validated on the same data. This allows direct comparison.

The goal of this thesis was to examine how well foundational time series models perform on emergency department occupancy prediction.

Five different foundational time series models were tested against trained machine learning models using same validation data. Different context lengths were tested. Giving multivariate data as a input was also tested.

2. BACKGROUND

2.1 Time series forecasting

- what is time series data
- how to forecasting is done (looking at historical data, capturing cyclical patterns)

2.2 Foundational models

- what foundational models are (trained using massive amounts of data etc)
- paradigm from ml model training+validation -> foundational models (saves computation, time and effort...)

2.3 Related work

2.3.1 Current state of art in ED forecasting

- tietoa nykyisestä state of artista ED ennustuksessa

2.3.2 Performance of foundational models in other fields

- Performance of foundational time series prediction for ED occ in other studies (if any)
- Performance of foundational time series prediction for other data (find studies)

3. METHODOLOGY

3.1 Models

- criteria for choosing models
- short and high level description of architectures and features of each model. naming of each model
- settings and hyper-parameters that were used for each model
- very short description of trained ml models that these models are compared against

3.2 Data

- where the data came from
- describe the data used (univariate, multivariate...) and how it was processed
- data allows direct comparison with other ml models trained on the same data.

3.3 Experimental setup

- task is to predict 24h occupancy to the future
- uni vs multivariate inputs
- different context length inputs
- hardware setup and (library versions, date when models were downloaded?)

3.4 Evaluation

- describe the evaluation metrics used (rmse, mae, msis), what they tell and why they are used

3.5 Finetuning

- tämä osio jos finetunausta tehdään

4. RESULTS

4.1 Main results

- show the charts and tables of the main results (rmse, mae, msis)

4.2 Effect of context length

- charts that show error with different context lengths

4.3 Effect of covariates

- chart comparing uni vs multi inputs

4.4 Visualizations

- show single day prediction vs ground truth. (best, worst and typical days)

5. DISCUSSION

- how accuracy changes with input length
- how input covariates changes accuracy
- did fine tuning increase accuracy
- are foundational models potential option for areas like this one, comparison to trained models, compare with other studies
- limitations of the study (only one validation dataset)
- other observations and future work
- maybe connect with results or conclusion chapter?

6. CONCLUSION

- summary of answers to the research questions, conclusion

7. REFERENCES